

Material Test Report / Certificate LSM-25-04412

Lone Star Specialty Metals, Inc. | 8800 Wallisville Rd, Houston, TX 77029

Certificate No. / Date	LSM-25-04412 / 2025-01-16
Heat / Batch No.	3M9077
Customer	American Completion Tools, Inc.
Purchase Order	ACT PO-87901
Product	Forged blocks (side pocket mandrel bodies), normalized, Q&T; to 75K
Specification(s)	AISI 4130 mod per ACT-MS-03; NACE MR0175/ISO 15156-2 (max 22 HRC)
Quantity	40 forgings / 11,300 lb (forging die F-SPM-287)

Chemical Composition (wt %)

Element	Value	Element	Value
C	0.30	Cr	0.95
Mn	0.52	Mo	0.18
P	0.012	Ni	0.06
S	0.009	Cu	0.12
Si	0.24	Fe	Bal.

Mechanical / Physical Properties

Property	Result	Requirement
Yield strength (0.2% offset)	79,800 psi	75,000 - 90,000 psi
Tensile strength	98,200 psi	>= 95,000 psi
Elongation in 2 in.	23 %	>= 18 %
Hardness	20 HRC	<= 22 HRC
Charpy V-notch @ -20 degF (avg of 3)	51 ft-lb	>= 25 ft-lb

Thermal Treatment

Normalized 1,650 degF; austenitized 1,600 degF, water quench; tempered 1,225 degF 4 hr

NDE / Additional Testing

WFMT per ASTM E709 post heat treat: no relevant indications

Sour Service (NACE MR0175 / ISO 15156) Statement

Certified compliant with NACE MR0175/ISO 15156-2 for sour service (<= 22 HRC, Q&T; condition).

Remarks

Certificate per EN 10204 Type 3.1. Used by ACT for: SPM-2875 side pocket mandrel bodies and welded pockets (weld per WPS-ACT-21), sour-service subs. Weld procedure qualification on file PQR-ACT-21A.

Certification

We hereby certify that the material described above was manufactured, sampled, tested and inspected in accordance with the referenced specifications and was found to meet those requirements. This certificate is issued in accordance with EN 10204 Type 3.1.

Role	Name	Signature	Date
Authorized Signatory	Q. Ramos, Chief Metallurgist, Lone Star Specialty Metals	/signed electronically/	2025-01-16
ACT Receiving QA Review	S. Nguyen	/signed electronically/	2025-01-16